# Tutorial 3

1. A wave of period, 10 s, propagates in a water depth of 12 m, the corresponding deep-water wave height $H_o = 3.5$ m. Estimate,

   **a.**

   Wave height

   **b.**

   Wave power

   **c.**

   the celerity using solitary wave theory

   **d.**

   the celerity of the wave using cnoidal wave theory for H=3.5m, H=3m

| $d/L_o$ | $d/L$ | $2\pi d/L$ | tanh ($2\pi d/L$) | sinh ($2\pi d/L$) | cosh ($2\pi d/L$) | $H/H'_o$ | K | $4\pi d/L$ | sinh ($4\pi d/L$) | cosh ($4\pi d/L$) | n | $C_G/C_o$ | M |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.0710 | 0.1149 | 0.7218 | 0.6180 | 0.7861 | 1.2720 | 0.9694 | 0.7862 | 1.4436 | 2.0000 | 2.2361 | 0.8609 | 0.5321 | 12.92 |
| 0.0720 | 0.1158 | 0.7277 | 0.6217 | 0.7937 | 1.2767 | 0.9676 | 0.7833 | 1.4554 | 2.0265 | 2.2598 | 0.8591 | 0.5341 | 12.77 |
| 0.0730 | 0.1168 | 0.7336 | 0.6253 | 0.8012 | 1.2814 | 0.9658 | 0.7804 | 1.4672 | 2.0532 | 2.2837 | 0.8573 | 0.5360 | 12.62 |
| 0.0740 | 0.1177 | 0.7394 | 0.6288 | 0.8087 | 1.2861 | 0.9641 | 0.7776 | 1.4789 | 2.0800 | 2.3079 | 0.8555 | 0.5379 | 12.48 |
| 0.0750 | 0.1186 | 0.7453 | 0.6323 | 0.8162 | 1.2908 | 0.9624 | 0.7747 | 1.4905 | 2.1071 | 2.3323 | 0.8537 | 0.5398 | 12.34 |
| 0.0760 | 0.1195 | 0.7511 | 0.6358 | 0.8237 | 1.2956 | 0.9608 | 0.7719 | 1.5021 | 2.1343 | 2.3570 | 0.8519 | 0.5416 | 12.21 |
| 0.0770 | 0.1205 | 0.7569 | 0.6392 | 0.8312 | 1.3004 | 0.9592 | 0.7690 | 1.5137 | 2.1618 | 2.3819 | 0.8501 | 0.5434 | 12.08 |
| 0.0780 | 0.1214 | 0.7626 | 0.6426 | 0.8387 | 1.3052 | 0.9577 | 0.7662 | 1.5253 | 2.1894 | 2.4070 | 0.8483 | 0.5452 | 11.95 |
| 0.0790 | 0.1223 | 0.7684 | 0.6460 | 0.8463 | 1.3100 | 0.9562 | 0.7633 | 1.5368 | 2.2173 | 2.4323 | 0.8466 | 0.5469 | 11.83 |
| 0.0800 | 0.1232 | 0.7741 | 0.6493 | 0.8538 | 1.3149 | 0.9547 | 0.7605 | 1.5483 | 2.2453 | 2.4579 | 0.8448 | 0.5485 | 11.70 |
| 0.0810 | 0.1241 | 0.7798 | 0.6526 | 0.8613 | 1.3198 | 0.9533 | 0.7577 | 1.5597 | 2.2736 | 2.4838 | 0.8430 | 0.5502 | 11.59 |
| 0.0820 | 0.1250 | 0.7855 | 0.6559 | 0.8689 | 1.3247 | 0.9519 | 0.7549 | 1.5711 | 2.3020 | 2.5099 | 0.8412 | 0.5518 | 11.47 |
| 0.0830 | 0.1259 | 0.7912 | 0.6591 | 0.8764 | 1.3297 | 0.9506 | 0.7521 | 1.5825 | 2.3307 | 2.5362 | 0.8395 | 0.5533 | 11.36 |
| 0.0840 | 0.1268 | 0.7969 | 0.6623 | 0.8840 | 1.3347 | 0.9493 | 0.7492 | 1.5938 | 2.3596 | 2.5628 | 0.8377 | 0.5548 | 11.25 |
| 0.0850 | 0.1277 | 0.8026 | 0.6655 | 0.8915 | 1.3397 | 0.9480 | 0.7464 | 1.6051 | 2.3888 | 2.5896 | 0.8360 | 0.5563 | 11.14 |
| 0.0860 | 0.1286 | 0.8082 | 0.6686 | 0.8991 | 1.3448 | 0.9468 | 0.7436 | 1.6164 | 2.4181 | 2.6168 | 0.8342 | 0.5578 | 11.04 |
| 0.0870 | 0.1295 | 0.8138 | 0.6717 | 0.9067 | 1.3498 | 0.9456 | 0.7408 | 1.6276 | 2.4477 | 2.6441 | 0.8325 | 0.5592 | 10.94 |
| 0.0880 | 0.1304 | 0.8194 | 0.6748 | 0.9143 | 1.3550 | 0.9444 | 0.7380 | 1.6389 | 2.4776 | 2.6718 | 0.8307 | 0.5606 | 10.84 |
| 0.0890 | 0.1313 | 0.8250 | 0.6778 | 0.9219 | 1.3601 | 0.9433 | 0.7352 | 1.6501 | 2.5076 | 2.6997 | 0.8290 | 0.5619 | 10.74 |
| 0.0900 | 0.1322 | 0.8306 | 0.6808 | 0.9295 | 1.3653 | 0.9422 | 0.7325 | 1.6612 | 2.5379 | 2.7278 | 0.8273 | 0.5632 | 10.65 |

**Solution**

Given data: $T = 10$ s and $H_o = 3.5$ m

$$L_o = 1.56T^2 = 1.56(10)^2 = 156 \text{ m},$$

$$d/L_o = 12/156 = 0.07692$$

From wave tables of SPM (1984),

$d/L = 0.1205$ and

hence $L = 99.58$ m $\qquad kd = 2\pi/L = 0.757$

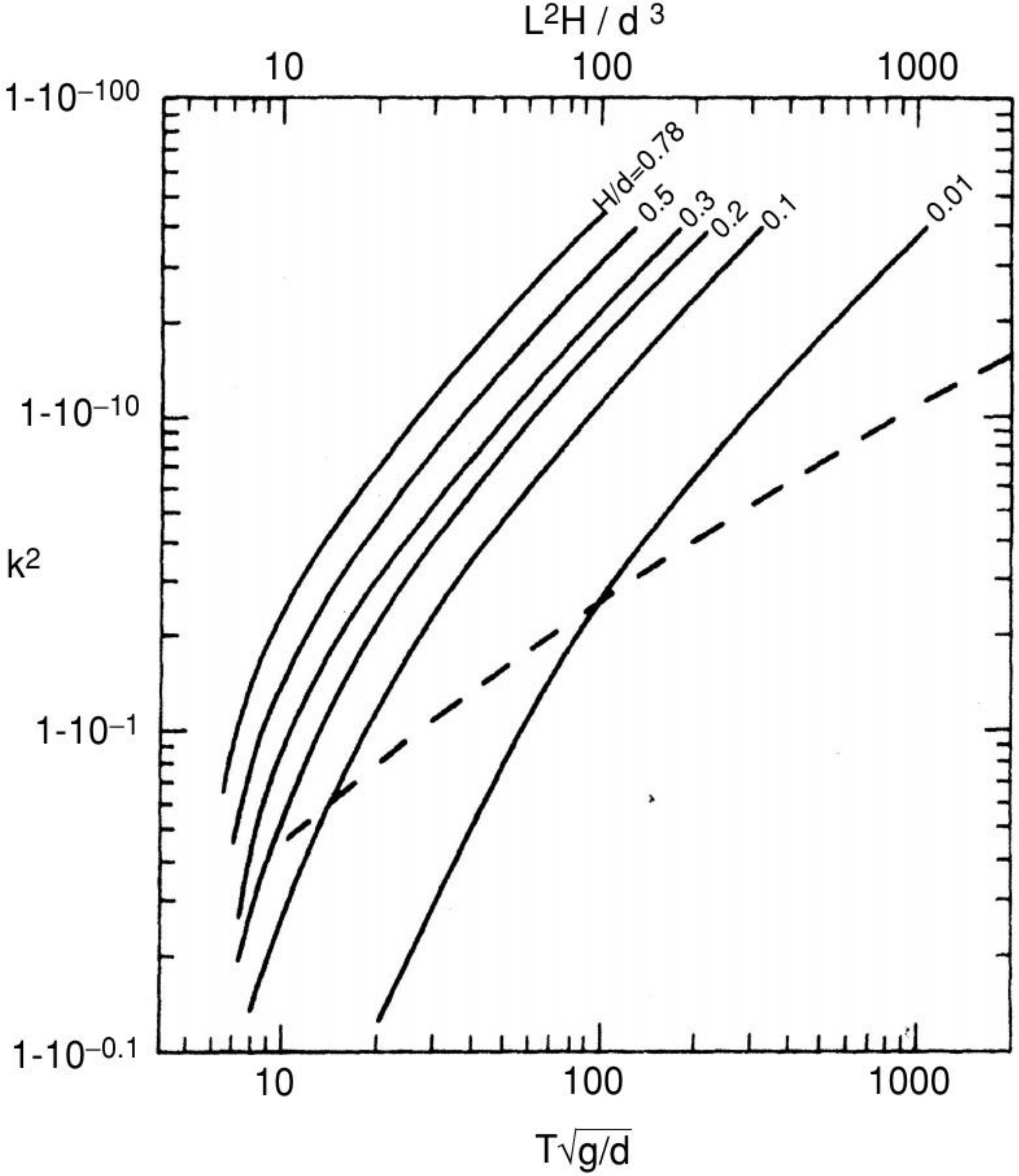
- a)
  - $C_o$=$L_o$/T=15.6m/s
  - C=L/T=9.958m/s
  - $\frac{H}{H_0}= \sqrt{\frac{C_0}{C} \cdot \frac{1}{2n}}$
  - $n = \frac{1}{2}\left[1 + \frac{2kd}{sinh2kd}\right]$=0.85
  - H=$H_o$*(15.6/(2*0.85*9.958))^0.5
  - H=$H_o$*0.96
  - H=3.36m

- b)

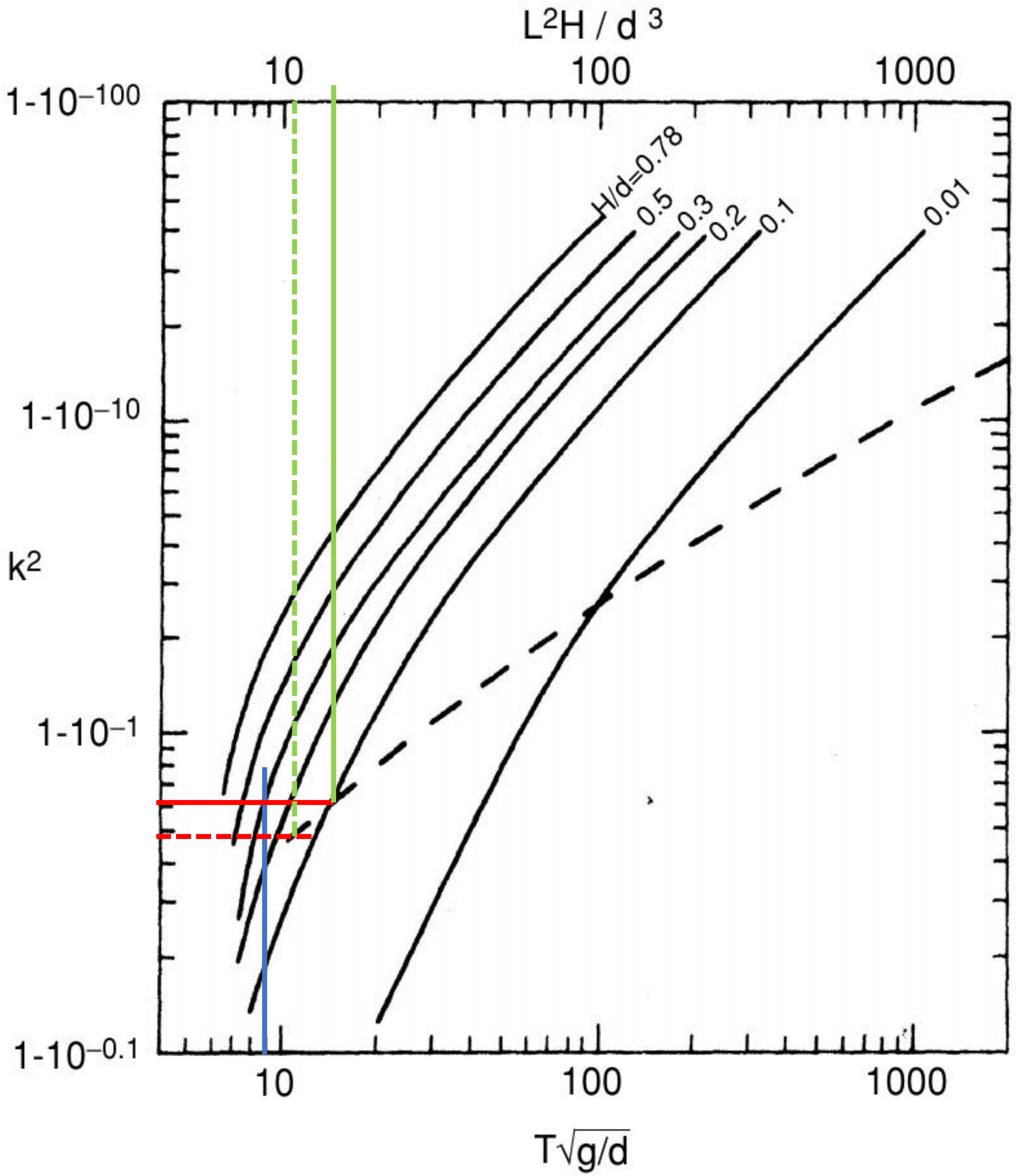
$$Wave\ power = \frac{\Upsilon H^2}{8}.n.C$$

**c.**

- $C = \sqrt{gd}\left(1 + \frac{H}{2d}\right)$
- $C = \sqrt{9.81 * 12}\left(1 + \frac{3.1}{2*12}\right) = 12.25 m/s$



L²H / d³
10
100
1000
1-10⁻¹⁰⁰
1-10⁻¹⁰
1-10⁻¹
1-10⁻⁰·¹
k²
H/d=0.78
0.5
0.3
0.2
0.1
0.01
10
100
1000
T√g/d




$$\frac{H}{d} = \frac{3.5}{12} = 0.3 \text{ (approx) (solid lines)}$$

and

$$T\sqrt{g/d} = 10\sqrt{9.81/12} = 9$$

This gives

$$U_r = 15 \text{ approx}$$

$$L = \sqrt{\frac{(12)^3 * 15}{3.5}} = 86 \text{ m}$$

$$C = 72/10 = 8.6 \text{ m/s}$$

$\frac{H}{d} = \frac{3}{12} = 0.25$ (since H/d curves are not demarcated in the chart, we vaguely select a region in the intermediate. Here the lines are dashed)

and

$$T\sqrt{g/d} = 10\sqrt{9.81/12} = 9$$

This gives

$$U_r = 11 \text{ approx}$$

$$L = \sqrt{\frac{(12)^3 * 11}{3}} = 79.6 \text{ m}$$

$$C = 72/10 = 7.96 \text{ m/s}$$